UNIVERSITY CAPSTONE · WEEK 11

W11

AfyaPlus Module 4 Integrated Capstone

Stub-first teaching demo · reviewable evidence · strict data boundaries

Who we serve

AfyaPlus Module 4 Integrated Capstone — Week 11

01 **Clinic coordinators and nurses in Kenya-style health operations (AfyaPlus)**

02 **One intake path: triage, logistics lookups, optional catalog checks**

03 **Strict boundary: clinical triage data stays separate from marketplace catalog data**

! **CLINICAL TRIAGE DATA ≠ MARKETPLACE CATALOG DATA**

AfyaPlus Module 4 Capstone

6–10 REUSE, NOT REBUILD

Integrated capstone
Official path for markers

Weeks 6–10 lab artefacts
—not a blank notebook

Teaching fixture: ~300 synthetic JSONL rows
—not a production scrape

W11 FINISHABLE MVP

Stub-first CI
One-week delivery

Finishable MVP in one week

Default stub mode keeps CI deterministic
and avoids live-key dependency

Chosen path: integrated capstone with bounded scope and honest fixtures.

System shape

POST /triage is the authenticated API boundary; agents/router selects a branch using keywords from config.json.

CLIENT → /TRIAGE → ROUTER
FastAPI · JWT
agents/router

BRANCH 1 · RAG
services/triage/
knowledge/*.md

BRANCH 2 · LOGISTICS
agents/tools/
logistics.py

BRANCH 3 · CATALOG
services/catalog/
stub_mismatch.py

Clinical triage fixtures and marketplace catalog fixtures remain separate.

Weeks 1–10 reuse

SKILL MATRIX · REUSED ARTEFACTS

6	WEEK 6 services/triage API
7	WEEK 7 agents/tools/logistics.py
8	WEEK 8 services/triage/prompts/pin.json
9	WEEK 9 tests/test_security_smoke.py · JWT · redaction
10	WEEK 10 Optional: services/catalog stub mismatch
11	WEEK 11 pytest pyramid · evals/golden.jsonl · doc pack

Week 11 closes the capstone with reviewable tests, evaluation data, and documentation.

3-step demo (port 8000)

01 Start: uvicorn services.triage.app:app --port 8000

02 GET /health → 200 with pin and version fields

03 POST /token for the nurse demo user

04 POST /triage with the Bearer token

RUN MODE

DEFAULT
stub_mode = true

OPTIONAL LOCAL LIVE
OPENAI_API_KEY set
FORCE_STUB = 0
gpt-4o-mini

CI always uses FORCE_STUB=1; no live key is stored or used in CI.

Reviewable system, not demo-only

Evidence is automated, repeatable, and inspectable.

01 **pytest -q unit layer green; integration marked separately**

02 **.env not tracked · no fake live keys · email redaction test**

03 **Golden eval: 3/3 routing score on evals/golden.jsonl**

04 **GitHub Actions: ruff, pytest, smoke, eval on every push/PR**

Week 9 habits carried forward

Four safeguards remain active in the capstone build.

JWT REQUIRED

401 without token

ENV HYGIENE

.env ignored

.env.example committed

FAIL CLOSED

No CI key → stub triage only

DATA BOUNDARY

No patient / marketplace fixture mixing

Teaching hygiene—not a compliance certification.

Cost analysis (teaching)

NO AUTONOMOUS AGENT LOOPS

- **A** — STUB TESTS / CI 0 live calls $0
- **B** — LOCAL STUB DEMO 0 live calls $0
- **C** — OPTIONAL LIVE DEMO ~10 mini calls teaching cap ~$1
- MODEL gpt-4o-mini
- LIMIT low max_tokens
- CONTROL local key + FORCE_STUB=0 only

The cap is an instructional guardrail, not a production forecast.

Build checkpoint

3 WORKS NOW

Reviewable MVP

MVP stub E2E

Route + RAG snippet + logistics tool

300-row fixture

fixtures/triage_messages_sample.jsonl

perf_stub.json teaching p95

2 DEFERRED

Documented next steps

Full Week 10 Sokoni mismatch.py pipeline

Production cloud deploy

Documented only

Honest boundary: perf_stub.json is a teaching p95, not an SLA.

Git, review, Week 12

01 Feature branch → PR #1 → self-review checklist → main

02 review_log.md updated; five docs linked from README

03 Week 12 Monday: present this deck + run live/stub demo

04 Close with the repository URL and questions

CLOSE

Sammy-CK/WEEK-11-CAPSTONE-PROJECT

Default to stub mode; use the optional live path only with a local key.